VARUNA — Sonar Mission Report

AI-powered automated underwater marine debris and anomaly detection

HIGH ALERT · Potential unexploded ordnance / mine-like object

Mission Details

Mission / XTF file	VARUNA_DEMO_MISSION.xtf
Frames processed	7
Total detections	6
Model	best.pt
Base confidence	0.25
NMS IoU	0.45
Navigation source	synthetic_demo
Geotag method	XTF ping navigation + sonar pixel geometry
Class thresholds	crab_pot: none, submarine_pipeline: 0.65, shipwreck: 0.55, ghost_net: 0.25, mine_cylinder: 0.35

Detection Summary

Class	Detections
submarine_pipeline	1
ghost_net	1
shipwreck	2
mine_cylinder	2

Geotagged Detections

#	Class	Confidence	Latitude	Longitude	Event	Depth (m)
1	submarine_pipeline	82.6%	15.396025	73.721333	3	35.0
2	ghost_net	95.9%	15.396425	73.722292	4	35.0
3	shipwreck	88.3%	15.396838	73.721840	5	35.0
4	shipwreck	60.0%	15.397940	73.723438	5	35.0
5	mine_cylinder	84.5%	15.396750	73.722896	6	35.0
6	mine_cylinder	80.3%	15.398289	73.723875	6	35.0

Detection Details

#	Class	Confidence	Event	Ping	Heading	Range (m)	Channel
1	submarine_pipeline	82.6%	3	1433	43.0°	18.31	port
2	ghost_net	95.9%	4	2185	47.0°	29.50	port
3	shipwreck	88.3%	5	2648	51.0°	41.89	port
4	shipwreck	60.0%	5	3187	51.0°	44.96	port
5	mine_cylinder	84.5%	6	3346	55.0°	43.33	port
6	mine_cylinder	80.3%	6	3816	55.0°	45.41	port

Annotated Sonar Frames

Frames containing accepted detections are included below.

Event 3

Annotated frame could not be embedded.

Event 4

Annotated frame could not be embedded.

Event 5

Annotated frame could not be embedded.

Event 6

Annotated frame could not be embedded.

Navigation / Geolocation Note

This demonstration mission uses synthetic XTF navigation data. Detection coordinates are derived from XTF ping navigation and sonar pixel geometry for demonstration purposes. They must not be interpreted as survey-grade real-world coordinates.

Generated by VARUNA — Offline AI Sonar Detection System (SIH 2026). This report is generated automatically from model output and should be verified by a qualified operator before action is taken.